

AI-based dental caries and tooth number detection in intraoral photos: Model development and performance evaluation

Kyubaek Yoon^a, Hye-Min Jeong^b, Jin-Woo Kim^c, Jung-Hyun Park^{c,*}, Jongeun Choi^{a,**}

^a School of Mechanical Engineering, Yonsei University, Seoul, South Korea

^b Department of Artificial Intelligence Convergence, Ewha Womans University, Seoul, South Korea

^c Department of Oral and Maxillofacial Surgery, College of Medicine, Ewha Womans University, Seoul, South Korea

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ABSTRACT

Objectives: In this study, we aimed to integrate tooth number recognition and caries detection in full intraoral photographic images using a cascade region-based deep convolutional neural network (R-CNN) model to facilitate the practical application of artificial intelligence (AI)-driven automatic caries detection in clinical practice. **Methods:** Our dataset comprised 24,578 images, encompassing 4787 upper occlusal, 4347 lower occlusal, 5230 right lateral, 5010 left lateral, and 5204 frontal views. In each intraoral image, tooth numbers and, when present, dental caries, including their location and stage, were annotated using bounding boxes. A cascade R-CNN model was used for dental caries detection and tooth number recognition within intraoral images.

Results: For tooth number recognition, the model achieved an average mean average precision (mAP) score of 0.880. In the task of dental caries detection, the model's average mAP score was 0.769, with individual scores spanning from 0.695 to 0.893.

Conclusions: The primary objective of integrating tooth number recognition and caries detection within full intraoral photographic images has been achieved by our deep learning model. The model's training on comprehensive intraoral datasets has demonstrated its potential for seamless clinical application.

Clinical Significance: This research holds clinical significance by achieving AI-driven automatic integration of tooth number recognition and caries detection in full intraoral images where multiple teeth are visible. It has the potential to promote the practical application of AI in real-life and clinical settings.

1. Introduction

In recent years, research on artificial intelligence (AI) integration in the field of dentistry, specifically in the domains of clinical diagnosis and treatment planning, has increased [1–4]. This can be attributed to the remarkable advancements in convolutional neural networks (CNNs), which have revolutionized the field of image analysis by leveraging deep learning techniques [5].

CNNs have become powerful tools in dentistry for image recognition and analysis, enabling the extraction of relevant features from medical images such as X-rays, CBCT scans, and MRIs. By leveraging their hierarchical structure, CNNs excel at learning complex patterns in dental images and capturing local and global features that are crucial for accurate diagnosis. This automated feature extraction enhances the

precision and efficiency of clinical decision-making, benefiting both dental professionals and patients through accurate diagnosis and treatment planning [6].

Most studies on automated caries detection using AI have focused on radiographic images [7–9]. Few studies have explored the application of RGB-based photographic images in this context [10–13]. However, these investigations predominantly involved manual segmentation of individual teeth within the image frame and the subsequent diagnosis of caries within each isolated tooth region. The approach employed in previous studies deviates from detecting actual caries in a patient's mouth because it failed to accurately replicate a typical oral environment [14]. Moreover, the strategy of identifying caries in isolated teeth necessitates an additional process and is constrained in amassing a wide range of diverse data. While there has been a study have addressed

* Corresponding author at: Department of Oral and Maxillofacial Surgery, College of Medicine, Ewha Womans University, 1071 Anyangcheon-ro, Yangcheon-gu, Seoul, 07985, South Korea.

** Corresponding author at: School of Mechanical Engineering, Yonsei University, 50 Yonsei Ro, Seodaemun Gu, Seoul, 03722, South Korea.

E-mail addresses: omspark07@gmail.com (J.-H. Park), jongeunchoi@yonsei.ac.kr (J. Choi).

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caries detection in full intraoral images, the focus has been solely on evaluating diagnostic accuracy, disregarding the crucial aspect of caries localization [15]. To enable the clinical application of automatic AI caries detection in intraoral images, it is imperative to identify and number each tooth in a full intraoral image and subsequently detect and localize dental caries in each tooth.

We hypothesized that by applying deep learning to full intraoral images, we could recognize individual tooth numbers and, if caries were present, simultaneously detect them. Therefore, this study aimed to integrate tooth recognition and caries detection using full intraoral images. To enhance the diagnostic accuracy and capabilities of caries diagnosis, we employed a three-stage classification system. Through this novel approach, we sought to provide a comprehensive assessment of caries detection, including precise tooth identification and the accurate localization of caries within the holistic context of a complete intraoral image.

2. Materials and methods

2.1. Data configuration

2.1.1. Intraoral photographic images

This study was approved by the Institutional Review Board of Ewha Medical Center, Seoul, Republic of Korea (IRB No. 2022-01-022-012). The intraoral images of patients who visited the hospital for dental treatment were collected retrospectively. The images were captured by dentists using professional DSLR (digital single-lens reflex) cameras after thoroughly cleaning and drying all the teeth. Images were captured from various angles, including occlusal, frontal, and lateral views. Mirrors were used for occlusal and lateral views, and an air syringe was employed to prevent condensation during the process. To ensure the highest image quality, the following images were excluded: 1) images with scratches on the mirror, if it was a mirror image; 2) images showing tooth surfaces obscured by saliva, food, orthodontic brackets, or other artifacts, which hinder a clear diagnosis of the tooth surface; 3) out-of-focus images; 4) images containing primary teeth, as primary teeth have different shapes from permanent teeth, which may interfere with AI learning for tooth recognition; and 5) images from the same patient showing the same tooth surface to prevent data duplication and ensure data integrity, guaranteeing that each clinical oral image was included only once. Only the regions with teeth were cropped and used in the dataset. Ultimately, 24,578 images were collected, comprising 4787 upper occlusal, 4347 lower occlusal, 5230 right lateral, 5010 left lateral, and 5204 frontal views.

The dataset was annotated by 20 data labelers who received training from a dentist (J-HP). They identified and labeled all the teeth in the intraoral images by assigning the corresponding tooth numbers using bounding boxes. To assist with the annotation process in our study, we crafted a specialized tool based on the open-source platform, Label Studio V1.0 (HumanSignal, San Francisco, California, USA). This tailored tool facilitates the precise demarcation of teeth within intraoral images, allowing labelers to create bounding boxes by dragging and dropping, which automatically records the coordinates of the box's upper-left corner, as well as its height and width. Utilizing this platform, labelers were able to identify and assign corresponding tooth numbers to all teeth, ensuring consistent and accurate labeling with bounding boxes.

Furthermore, if present, dental caries were annotated, indicating their location and stage based on the American Dental Association Caries Classification System (ADA CCS) and International Caries Detection and Assessment System (ICDAS) [16,17]. Stage 1 corresponded to a 'moderate caries lesion' in ADA CCS, equivalent to ICDAS codes 3 (localized enamel breakdown because of caries with no visible dentine or underlying shadow) and code 4 (underlying dark shadow from dentine with or without localized enamel breakdown without cavitation into dentine). Stage 2 corresponded to an 'advanced caries

lesion' in ADA CCS, equivalent to ICDAS Codes 5 (distinct cavity with visible dentine) and code 6 (Extensive - more than half the surface - distinct cavity with visible dentine). Stage 3 signified complete destruction, with minimal residual structure, representing teeth in a fully compromised state. This inclusion was intended to enhance the potential of AI's visual detection of tooth remnants.

To ensure precision and consistency for dental caries staging, the annotation process underwent rigorous examination by three dentists. They received theoretical and practical calibration training focused on the staging of dental caries, guided by J-H P. The theoretical training covered study objectives and diagnostic criteria. The practical component included the analysis and discussion of numerous photographs of dental caries lesions with possible differential diagnosis [18]. At the end of the calibration training, each dentist evaluated 368 unknown photographs to detect dental caries lesions and staging. This sample size was determined considering the expected agreement based on previous studies, the distribution of caries stages, and the number of examiners. To assess the inter-examiner reliability in dental caries staging, Kappa values were calculated, resulting in an inter-examiner reproducibility of $\kappa = 0.744$ (95 % confidence interval, 0.701–0.787), indicating substantial agreement [19].

2.1.2. Dataset

The dataset comprised intraoral images and the corresponding JavaScript Object Notation (JSON) files [20]. The JSON file encapsulated pertinent information regarding the spatial positioning of teeth and cavities in each intraoral image. These data include the coordinates of the bounding boxes that delineate the respective regions of interest, along with tooth numbering and dental caries stage details. The JSON file adhered to the standardized format employed for the MS COCO dataset [21]. Supplementary Tables 1 and 2 provide a comprehensive overview of the instance counts per class, offering a concise summary of the frequency of each class within the dataset.

2.1.3. Data splitting

In this study, we carefully determined the data-splitting ratio to effectively allocate samples to the training, validation, and test sets. Specifically, we adopted a ratio of 8:1:1 to ensure a sufficiently large training set while providing ample data for validation and reliable testing. Further explanations regarding data-splitting process are detailed in the Supplementary Method 1.

2.2. Deep learning model for object detection

Our study explored the application of two distinct models, each employing a cascade region-based convolutional neural network (cascade R-CNN) [22] framework, to address separate tasks within intraoral images: one for dental caries detection and another for tooth number recognition.

The caries detection model detected and localized regions within an image that exhibited signs of caries. These regions were represented by bounding boxes that provided spatial information regarding the location and extent of detected caries. Furthermore, the model classified the severity of caries into three stages: 1, 2, and 3. This classification scheme enabled a fine-grained assessment of the carious areas.

The tooth number recognition model also identified and localized the bounding boxes for all tooth present in the image. By recognizing the number of teeth associated with each detected tooth, we inferred the number of teeth affected by caries.

In the development of our model, we employed Python [23] for data preprocessing and statistical evaluations, utilizing libraries that are well-documented and standard within the field. For model training, we utilized the PyTorch [24] framework, which is noted for its flexibility, efficiency, and suitability for deep learning tasks. In conducting our research, the hardware setup consisted of a dedicated computational workstation equipped with Intel Xeon E5-2698 processors, 256GB of

DDR4 RAM, and two NVIDIA Tesla V100 GPUs [25] with 32GB of VRAM each from NVIDIA Corporation, Santa Clara, California, USA. This system was chosen for its ability to efficiently manage the intensive computational demands of deep learning algorithms and large-scale data processing involved in our model's training and evaluation phases.

2.2.1. Model architecture

The overall structure of the cascade R-CNN model consists of a backbone network, a region proposal network (RPN), and a region of interest (RoI) head. The cascade R-CNN model is equipped with a ResNet-100 backbone network that acts as a feature extractor, vital for identifying relevant features from intraoral photographic images. This backbone, with a depth of 100 layers, leverages a pre-trained model to facilitate robust feature extraction.

Region proposals are generated by the RPN, which employs an Anchor Generator. The Anchor Generator is configured with scales of 8.0 and aspect ratios of 0.5, 1.0, and 2.0, allowing the model to propose candidate regions across various object sizes, a necessary consideration given the diverse range of tooth sizes and shapes.

For each proposed region, the RoI head applies a cascade structure with three stages, each utilizing a Shared2FCBBoxHead. This shared bounding box head refines the proposals through each stage. The Shared2FCBBoxHead, integral to the model's architecture, enables a consistent feature extraction process across multiple stages, contributing to the cascade effect of sequentially improving the precision of object detection.

Bounding box regression is performed using a DeltaXYWHBBoxCoder at each stage of the RoI head. This coder applies delta transformations to adjust the scale and position of the proposed bounding boxes, improving their alignment with the ground truth. The target standard deviations for the bounding box regression—[0.1, 0.1, 0.2, 0.2] for the initial stage, [0.05, 0.05, 0.1, 0.1] for the intermediate, and [0.033, 0.033, 0.067, 0.067] for the final stage—demonstrate a progressive approach to achieving precise localization accuracy.

This sophisticated cascade R-CNN architecture, with its methodical parameter tuning and the inclusion of an Anchor Generator, Shared2FCBBoxHead, and DeltaXYWHBBoxCoder, ensures the model's efficiency and efficacy. By meticulously refining the detection and classification process across stages, the model proves to be highly capable for the nuanced task of caries detection and tooth number recognition within the complex visual field of full intraoral photographic images.

2.3. Model training

During the model training phase, the following steps were performed:

1. **Dataset Preparation:** The training dataset was loaded, which included intraoral images and the corresponding annotations with bounding boxes for dental caries and tooth numbers. The annotations provided ground-truth information for training the model.
2. **Data transformation:** For the tooth numbering recognition model, augmentations were performed to foster model robustness and generalizability. These transformations included resizing the images to a uniform scale of 1333×800 pixels to standardize input dimensions, applying random flipping to introduce variation, normalizing the image pixel values to ensure consistent input ranges for the neural network, and adding padding as necessary to maintain compatibility with the model's architecture. The caries detection model, on the other hand, employed a more extensive set of augmentations to capture the variability of carious lesions. The images were resized to the same scale of 1333×800 pixels while maintaining the aspect ratio to preserve the integrity of the carious indicators. Random flipping was utilized to simulate different orientations of caries presentation. A random shift operation was

included to simulate positional variances. Photometric distortions were applied to simulate changes in lighting conditions and image contrast, with a brightness delta of 16 and a contrast range between 0.7 and 1.3. Furthermore, a cut-out augmentation was used to introduce occlusion, creating up to five holes per image with sizes ranging from 50×50 to 60×60 pixels and filling them with black (0, 0, 0), to train the model in identifying caries even in partially obscured views.

3. **Model Initialization:** The cascade R-CNN model was initialized from a pretrained model trained on the COCO dataset[21] using the ResNet-101 [26] backbone network and FPN structure. Starting with these pretrained weights, our model's effectiveness in dental caries detection and tooth number recognition was enhanced by the large dataset.
4. **Optimization Setup:** The parameters of the model were optimized using stochastic gradient descent with a learning rate of 0.0002 and a weight decay of 0.0000001. The learning rate followed a step-based policy with warm-up and decay phases. The optimizer aims to minimize the loss function during training.
5. **Training Loop:** The model underwent an iterative training loop over multiple epochs using batches of images and annotations. Each epoch represented a complete pass through the training dataset. For model training, we utilized two Tesla V100 GPUs [25] from NVIDIA Corporation, Santa Clara, California, USA. The operating system was Ubuntu 20.04 LTS, which provided a stable and reliable operating system environment using the Linux kernel and based on Debian for all computational tasks. The batch size was eight samples per GPU, and the training process lasted for 40 epochs. In each iteration, the model computed the forward pass, predicted the bounding box locations and class labels for dental caries and tooth numbers, and compared them with the ground-truth annotations. The model parameters were updated based on the losses calculated using backpropagation. This iterative process refines the ability of the model to accurately detect dental caries and recognize tooth numbers in intraoral images.
6. **Loss Calculation and Backpropagation:** The loss was calculated by comparing the predicted results with the ground-truth annotations. The losses were computed separately for the RPN and RoI heads. The gradients were then backpropagated through the network, and the parameters of the model were updated to improve its performance.
7. **Evaluation:** The model's performance was periodically evaluated using a validation dataset. The model predictions were compared with ground-truth annotations to calculate evaluation metrics such as the mean average precision (mAP). These metrics indicate the accuracy of the model and help monitor its progress during training.
8. **Saving Checkpoints:** Checkpoints in the model were saved at regular intervals, allowing the training process to be resumed or evaluated later. The best-performing model based on the evaluation metric (e.g., highest mAP) was typically saved for the model test phase.

After training the model, it underwent a testing phase (Supplementary Method 2).

2.3.1. Evaluation metrics

Performance was evaluated using mAP scores with an intersection over union (IoU) threshold of 0.5, which is a standard metric for object detection in deep learning. Diagnostic performance was determined by calculating sensitivity, specificity, accuracy, and the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC). A detailed description of these evaluation methods is available in Supplementary Method 3.

3. Results

3.1. Overall performance

The tooth number recognition model demonstrated an overall mAP of 0.880, with the highest individual tooth mAP recorded for tooth 21 at 0.975 and the lowest for tooth 48 at 0.372, as reported in Table 1. In the task of caries detection, the model achieved an overall mAP of 0.769. As shown in Table 2, the model detected Stage 3 caries with an mAP of 0.893. Stage 1 and Stage 2 caries were detected with mAPs of 0.718 and 0.695, respectively.

Fig. 1 presents the Receiver Operating Characteristic (ROC) curves of caries detection and tooth recognition models, and Fig. 2 presents mean average precision changes over epochs in model validation, Fig. 3 presents the visualization results from multiple views of the intraoral photographic images, illustrating both tooth number recognition and dental caries detection.

3.2. Performance in frontal views

For the frontal view in tooth number recognition, the incisors such as teeth 11 and 21 presented the highest mAP scores, 0.991 and 0.990 respectively, underscoring the model’s effectiveness in these perspectives. The model’s capacity to discern posterior teeth in the frontal view, however, was less precise. In caries detection from frontal views, Stage 3 lesions were detected with an mAP of 0.984. The mAP for Stage 1 lesions stood at 0.580.

3.3. Performance in upper and lower occlusal views

The tooth number recognition model excelled in the upper occlusal view, with teeth 11, 12, and 13 achieving high mAP scores of 0.992, 0.989, and 0.991 respectively. For the lower occlusal view, the model achieved an mAP of 0.980 with tooth 44. In the caries detection model, the upper occlusal view showed an mAP of 0.817 for Stage 3, and an mAP of 0.812 for Stage 1.

3.4. Performance in right and left lateral views

Teeth 16 and 46 had the mAP scores of 0.983 and 0.977, respectively. The wisdom teeth, such as 48 and 38, had the mAP scores of 0.469 and 0.422, respectively. For dental caries detection, Stage 3 lesions in the left lateral view were detected with an mAP of 1.00.

Table 1
Mean average precision scores for tooth number recognition.

Tooth	Overall	Frontal view	Occlusal view	Lateral view	Tooth	Overall	Frontal view	Occlusal view	Lateral view
11	0.974	0.991	0.992	0.950	21	0.975	0.990	0.992	0.952
12	0.969	0.990	0.989	0.962	22	0.974	0.989	0.992	0.964
13	0.971	0.990	0.991	0.959	23	0.966	0.984	0.988	0.966
14	0.965	0.988	0.990	0.953	24	0.969	0.986	0.990	0.966
15	0.947	0.985	0.982	0.955	25	0.948	0.980	0.984	0.957
16	0.938	0.950	0.983	0.960	26	0.946	0.981	0.979	0.955
17	0.930	0.948	0.951	0.953	27	0.906	0.922	0.935	0.909
18	0.555	0.452	0.663	0.598	28	0.573	None	0.714	0.582
41	0.916	0.940	0.947	0.929	31	0.924	0.950	0.952	0.937
42	0.924	0.954	0.957	0.940	32	0.932	0.951	0.950	0.940
43	0.943	0.962	0.979	0.951	33	0.939	0.963	0.967	0.959
44	0.939	0.953	0.980	0.965	34	0.948	0.979	0.982	0.958
45	0.921	0.947	0.950	0.941	35	0.932	0.949	0.961	0.944
46	0.939	0.959	0.977	0.960	36	0.938	0.952	0.970	0.943
47	0.914	0.939	0.930	0.928	37	0.890	0.910	0.929	0.891
48	0.372	0.438	0.686	0.469	38	0.294	0.407	0.628	0.422

Note. The values depicted in the cells of the table correspond to the mean average precision (mAP) scores.

Table 2
Mean average precision scores for caries detection.

Caries stage	Overall	Frontal view	Upper occlusal view	Lower occlusal view	Right lateral view	Left lateral view
Stage 1	0.718	0.580	0.812	0.729	0.624	0.609
Stage 2	0.695	0.620	0.733	0.707	0.658	0.615
Stage 3	0.893	0.984	0.817	0.836	None	1.00

Note. The values depicted in the cells of the table correspond to the mean average precision (mAP) scores.

3.5. Diagnostic performance

The diagnostic performance for tooth number recognition, detailed in Table 3, shows an average sensitivity of 0.775, with specific teeth ranging from 0.773 to 0.799. Specificity for all teeth is at 0.992 on average, with a range from 0.992 to 0.999. The accuracy of tooth recognition is 0.985 on average, with values for Teeth 48 and 38 up to 0.999. The Area Under the Curve (AUC) for the ROC curves has an average of 0.916, with values for Tooth 28 up to 0.979.

Table 4 reports the diagnostic performance for caries detection, with an average sensitivity of 0.732 and a range from 0.706 to 0.781. The average specificity is reported at 0.968, with a range up to 0.989. Caries detection accuracy averages at 0.950, with Stage 3 achieving up to 0.977. The average AUC is 0.943, with Stage 3 reaching up to 0.965.

4. Discussion

This study focused on the detection of dental caries and tooth number recognition using intraoral photographs. Our cascade R-CNN model showed promising results. The model demonstrated effective detection and recognition capabilities, contributing to a comprehensive understanding of tooth conditions. Our performance evaluation demonstrated high mAP scores for caries detection and tooth number recognition using our model. Specifically, the model achieved AP scores of 0.718, 0.695, and 0.893 for caries Stages 1 (enamel breakdown or dark shadow from dentin), 2 (distinct cavity), and 3 (extensive crown destruction), respectively, demonstrating its ability to accurately identify and localize different stages of dental caries. Additionally, the model exhibited a strong performance in tooth number recognition, with notable AP scores of 0.975 for tooth 21, 0.974 for tooth 22, and 0.974 for tooth 11. These results highlighted the proficiency of the model in recognizing specific tooth numbers in intraoral images.

The deep-learning model demonstrated an intriguing capability for

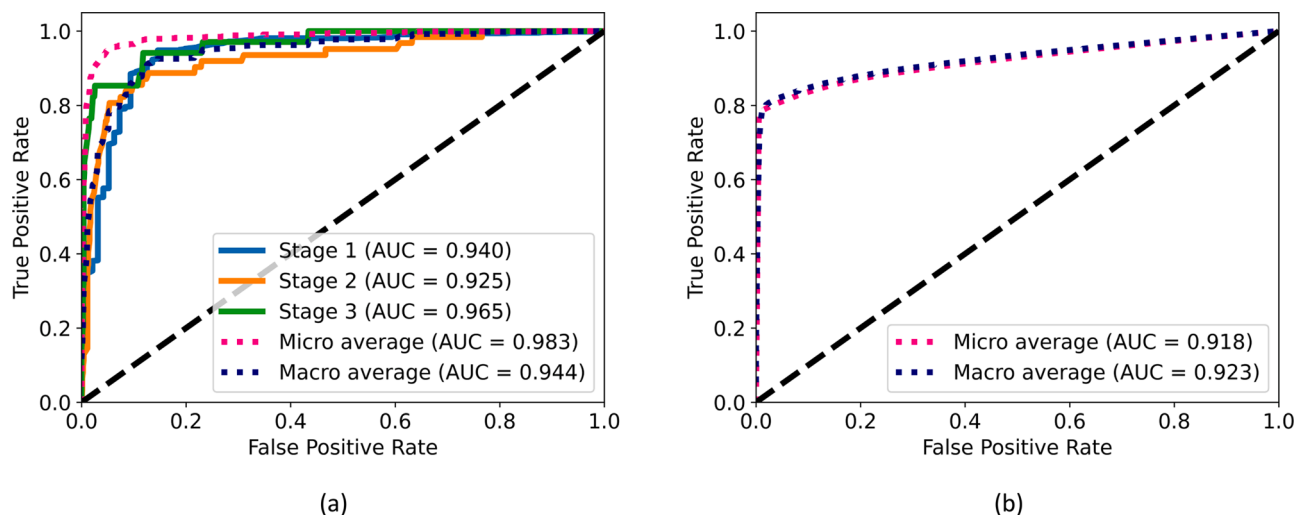


Fig. 1. Receiver operating characteristic (ROC) curves for caries detection and tooth number recognition. (a) ROC curves for dental caries detection across different stages. (b) ROC curves for tooth number recognition.

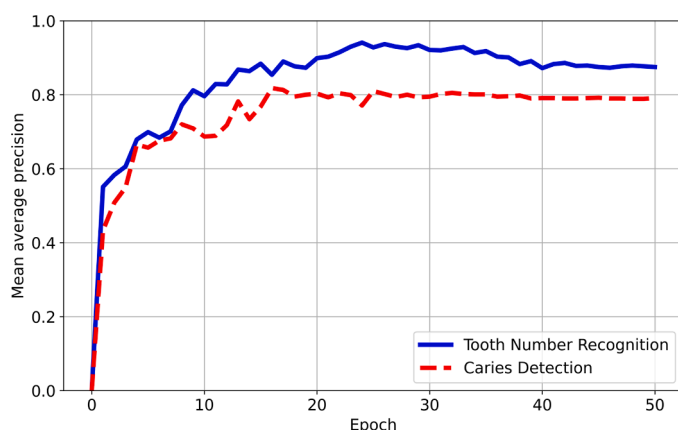


Fig. 2. Mean average precision (mAP) across training epochs for tooth number recognition and caries detection models in model validation.

tooth number recognition. Although the dental structures were inherently symmetrical from left to right, the model efficiently differentiated between them. This suggests that deep learning models are adept at recognizing the nuances of dental structures, even when faced with their symmetrical nature. The capacity of a model to discern local features among global symmetry points depends on the strength and sensitivity of its architecture. However, the performance for some tooth classes, specifically 18, 28, 38, and 48, wasn't as strong as others. This could be due to the limited data available for these classes compared to others. This emphasizes the need for a comprehensive dataset in training deep learning models, as data richness greatly impacts a model's ability to differentiate and categorize.

By switching gears for dental caries detection, interesting observations were made. Despite having fewer training images for Stage 3 than for Stage 1, the model performed better at detecting Stage 3 caries. A possible interpretation of this phenomenon is rooted in the visual characteristics of the caries at different stages. Stage 3 caries tended to exhibit more pronounced external features in intraoral images, making them more discernible. In contrast, the subtle nature of stage 1 caries, given its lower intensity, may pose challenges to the deep learning model. This suggests that, while the quantity of data is paramount, the qualitative features of the data, especially those that are distinctly discernible, play a crucial role in the model's performance.

The diagnostic capabilities of the tooth number recognition and

caries detection models were quantified by calculating sensitivity, specificity, accuracy, and the AUC relative to ground truth boxes. These metrics extend beyond the AP metric, which does not account for predictions below a certain confidence threshold, to offer a full spectrum evaluation of the model's performance. The analysis revealed that teeth 18, 28, 38, and 48 yielded diagnostic performance metrics within the range observed for other teeth, suggesting that the model's predictive ability, once teeth are accurately localized, does not vary significantly among different tooth groups. For caries detection, AUC scores exceeded the 0.9 threshold, indicating a consistent model performance in staging carious lesions.

Previous studies on caries detection in intraoral photographic images have primarily focused on models using single-segmented tooth images or images of extracted teeth. These isolated tooth studies have the advantage of enhancing diagnostic accuracy by providing precise surface details through high-resolution tooth images. However, when capturing intraoral photographs, it is impossible to capture images that include only one tooth. Instead, several teeth were captured from various directions, highlighting the need for validating the aforementioned laboratory-level model using images encountered in real clinical practice. Thanh et al. proposed a deep learning model for dental caries diagnosis using full intraoral images from smartphone intraoral photographs [15]. However, their model primarily evaluated the diagnostic accuracy of the binary classification and did not offer precise outputs for caries localization in full intraoral images.

In contrast to previous studies, the primary goal of this study is to assess the accuracy of caries diagnosis and tooth numbering in all teeth from various perspectives, thereby providing valuable insights for developing a more precise caries diagnosis model using full intraoral images. By identifying areas that require additional learning to improve accuracy, we expect increased efficiency through targeted data collection for these specific regions. A model capable of identifying specific areas of teeth affected by caries within full intraoral images, obtainable during routine clinical practice or in everyday life situations, is well suited for both clinical practice and daily life applications.

The success of the proposed model can be attributed to the effectiveness of the cascade R-CNN architecture. By combining R-CNNs with a cascade structure, the model progressively refines object detection and classification. Compared to the YOLOv3, Faster R-CNN, RetinaNet, and SSD models employed in [15], our model presents distinct advantages. The cascade R-CNN utilizes a multistage approach to progressively refine object detection and enhance precision. This hierarchical architecture allows for iterative filtering of false positives and focuses on more challenging instances, ultimately leading to higher accuracy in

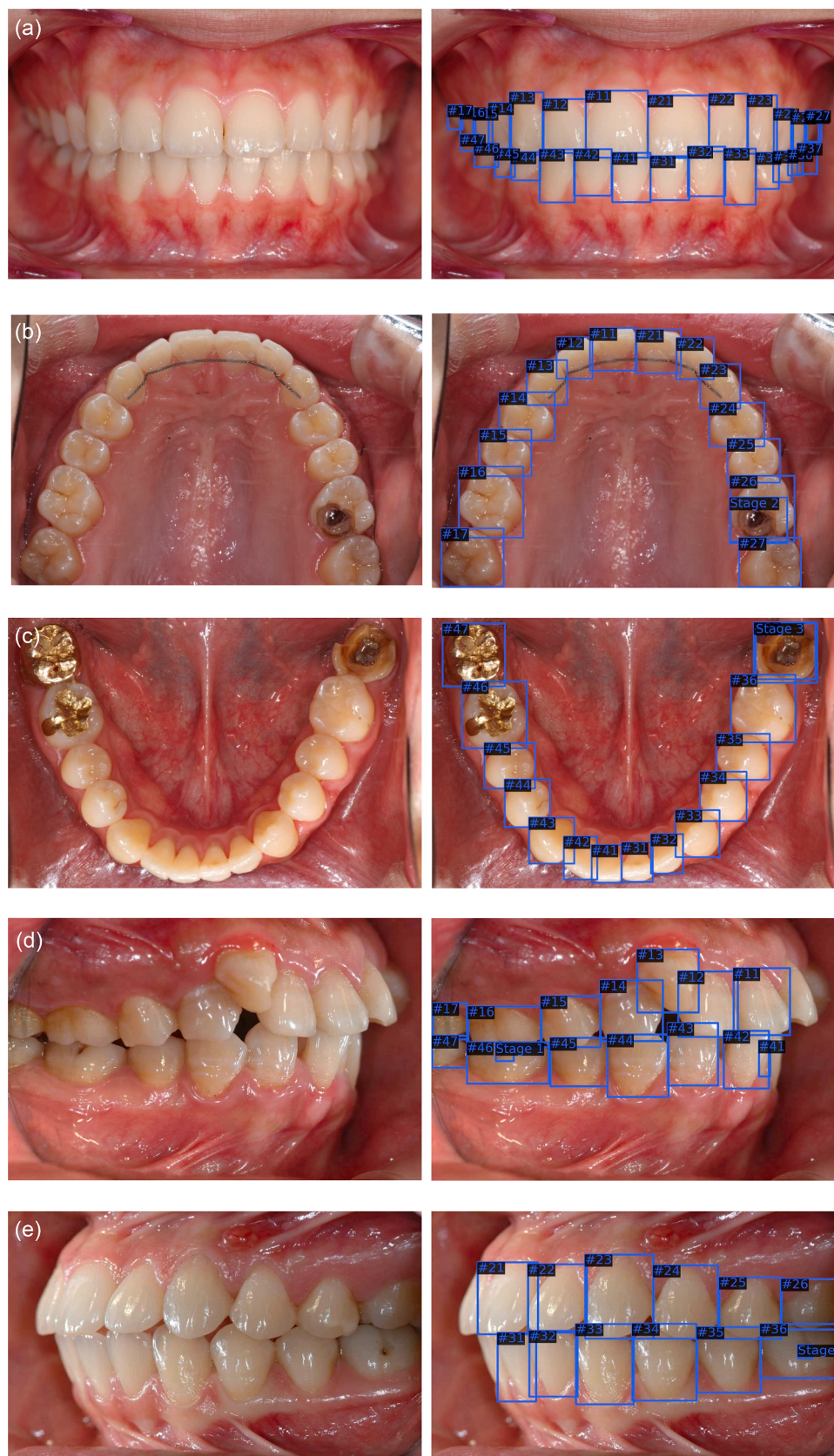


Fig. 3. Caries detection and tooth numbering recognition in intraoral photographic images. For each view, the left image is the original input, while the right showcases the model's output with objects detected by bounding boxes. (a) Frontal view. (b) Upper occlusal view. (c) Lower occlusal view. (d) Right lateral view. (e) Left lateral view.

Table 3
Diagnostic performance metrics for tooth number recognition.

Tooth	Sensitivity	Specificity	Accuracy	AUC	Tooth	Sensitivity	Specificity	Accuracy	AUC
11	0.773	0.991	0.982	0.908	21	0.770	0.993	0.982	0.902
12	0.774	0.986	0.979	0.903	22	0.786	0.989	0.981	0.909
13	0.760	0.992	0.984	0.902	23	0.775	0.992	0.983	0.910
14	0.771	0.991	0.984	0.909	24	0.772	0.992	0.982	0.909
15	0.795	0.990	0.986	0.930	25	0.782	0.993	0.986	0.918
16	0.785	0.992	0.987	0.947	26	0.778	0.994	0.986	0.930
17	0.799	0.996	0.994	0.960	27	0.773	0.997	0.993	0.942
18	0.715	0.999	0.999	0.958	28	0.788	0.999	0.999	0.979
41	0.778	0.983	0.973	0.908	31	0.770	0.989	0.977	0.909
42	0.773	0.989	0.980	0.904	32	0.783	0.990	0.978	0.912
43	0.765	0.992	0.983	0.897	33	0.775	0.992	0.981	0.906
44	0.766	0.991	0.984	0.903	34	0.777	0.991	0.982	0.905
45	0.778	0.993	0.987	0.916	35	0.774	0.994	0.985	0.909
46	0.774	0.993	0.987	0.929	36	0.770	0.993	0.982	0.914
47	0.813	0.994	0.992	0.950	37	0.761	0.996	0.990	0.922
48	0.764	0.999	0.999	0.958	38	0.769	0.999	0.999	0.992

Table 4
Diagnostic performance metrics for caries detection.

Caries stage	Sensitivity	Specificity	Accuracy	AUC
Stage 1	0.781	0.956	0.934	0.940
Stage 2	0.710	0.959	0.939	0.925
Stage 3	0.706	0.989	0.977	0.965

dental caries detection and tooth number recognition. The integration of the cascade R-CNN architecture has proven to be instrumental in enhancing the model’s performance and addressing the complexities of dental image analysis tasks.

Large datasets are crucial for improving deep-learning model performance. Including many diverse images helps the model learn effectively, enhancing accuracy in dental caries detection. Kühnisch et al. [12] stress the need for thousands of images representing all teeth, surfaces, and caries types to overcome data saturation. Our study used large datasets for training; however, only 1938 images (3521 teeth) of 24,578 contained dental caries. Additionally, more teeth had no cavities (Stage 1) than cavities (Stages 2 and 3). This data imbalance can bias metrics, limiting CNN training and caries detection performance. It emphasizes the necessity to continually expand the dataset, enhancing the model’s generalizability and efficiency in detecting various carious lesions on teeth and surfaces.

The deployment of our models on a web application could herald a new era of accessibility in dental diagnostics, particularly for self-assessment of caries. This advancement holds the promise of transforming the landscape of dental health monitoring, by empowering individuals with the tools to perform preliminary evaluations of their dental health. The integration of these models into a web-based platform necessitates not only a rigorous process of software optimization to ensure efficiency and responsiveness but also a robust hardware infrastructure that can support the computational demands of deep learning algorithms. Preparatory steps will include the lightweight redesign of our models to optimize for speed and lower resource consumption, ensuring they are suitable for diverse internet conditions and user devices. Concurrently, server capabilities must be enhanced to facilitate high-traffic usage and secure storage of potentially sensitive health data.

This study suggests that the scope of our model extends beyond caries detection to potentially include other oral diseases, such as periodontitis. Additionally, there is potential for expansion in dental prosthetic detection. In future research, it is imperative to gather data to facilitate these extensions. We aim to develop a deep learning model capable of identifying a diverse range of oral diseases and dental prosthetics by improving the performance.

5. Conclusion

In conclusion, our deep learning model effectively detected caries and recognized the tooth numbers in intraoral photographic images. Our model, which was trained on full intraoral images, is poised for effective clinical integration. Future research is needed to expand the model’s capabilities to diagnose other oral diseases such as periodontitis and detect dental prosthetics, aiming for effective integrated clinical utilization.

CRedit authorship contribution statement

Kyubaek Yoon: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Writing – original draft. **Hye-Min Jeong:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Jin-Woo Kim:** Data curation, Formal analysis, Funding acquisition, Writing – review & editing. **Jung-Hyun Park:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Jongueun Choi:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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